2024 IEEE-RAS 23rd International Conference on Humanoid Robots (Humanoids)
Nancy, France. November 22-24, 2024

2024 IEEE-RAS 23rd International Conference on Humanoid Robots (Humanoids) | 979-8-3503-7357-8/24/$31.00 ©2024 IEEE | DOI: 10.1109/Humanoids58906.2024.10769797

Therapist's and Robot's Roles in Robot-Assisted Interventions during JA Therapies of Children with ASD

Andrés Aguirre[1], Sergio D. Sierra M.[2], Maria Gaitán-Padilla[3], Anselmo Frizera-Neto[3], Carlos A. Cifuentes[2], Marcela Múnera[2,1]

Abstract— **Joint Attention (JA) is a fundamental social interaction capability developed in early childhood by sharing a common focus point with others. However, this skill is commonly affected in children with Autism Spectrum Disorders (ASD). In this sense, social robots emerged as a tool for developing novel JA intervention strategies supporting therapists. Therefore, this work presents a study to determine the best combination of robot and therapist participation in JA therapies based on following the gaze of the mediator with verbal and pointing instructions. Sixteen subjects with ASD participated in seven robot-assisted sessions divided into a robot-assisted group (RAG) and a control group (CG), performing equivalent intervention sessions. A focus visual system measured quantitative and reliable JA metrics, while the robot's participation was progressively increased in the RAG to manage all the instructions at the last session. Results show that participants of the RAG had better JA scores than the CG during sessions 6 and 7 ($p = 0.029$ and $p = 0.018$, respectively). The RAG paid more visual attention and presented more social engagement behaviours in the sessions ($p < 0.05$). Moreover, the best JA performance was given when the therapist and the robot requested similar instructions during the activities.**

I. INTRODUCTION

According to the Centre for Disease and Control Prevention, about 1 in 59 children exhibit Autism Spectrum Disorder (ASD) [1]. ASD is a lifelong neuro-developmental condition that affects the social and communication skills of children [2]. People with ASD often exhibit persistent deficits at interacting with people [2], getting engaged in active tasks [3], recognizing emotional states [4], learning new concepts[1], maintaining eye contact [1], and interpreting gaze as a social cue [4].

Joint Attention (JA) is the skill based on the capacity to share visual and cognitive attention with a partner to an external target. It commonly develops during the first 18 months of life [1]. Face-to-face visual engagement is an initial sign of JA development during this period. Over six months old, sharing visual attention to objects evolves as a passive capability. During the next months, other behaviours might show active engagement with others (e.g., external focusing, talking, target pointing, touching, and looking back to a partner) [5].

This work was supported in part by the Royal Academy of Engineering, CASTOR Project: CompliAnt SofT Robotics (Grant IAPP1-100126) and Minciencias Colombia (Grant 845-2020)

[1] Colombian School of Engineering Julio Garavito, Bogotá D.C., Colombia. andres.aguirre@mail.escuelaing.edu.co

[2] Bristol Robotics Laboratory, University of the West of England, Bristol, UK. sergio.sierra@uwe.ac.uk, marcela.munera@uwe.ac.uk, carlos.cifuentes@uwe.ac.uk

[3] Graduate Program of Electrical Engineering, Federal University of Espirito Santo, Vitoria 29075-910, Brazil. maria.padilla@edu.ufes.br, anselmo.frizera-neto@ufes.br

Several studies focus on designing and assessing robot-assisted protocols for JA screening [5]. For instance, RGB-depth cameras and facial recognition systems can quantify JA indicators and obtain more accurate diagnoses [6]. These systems also stimulate multiple responses in children. The clinical community has recognized robot-assisted diagnosis as an appropriate tool for diagnosing ASD [5], [7]. A recent type of therapeutic intervention is developed using Socially Assistive Robotics (SAR) [8]. Several studies reported that children with ASD are more attracted to interacting with technological systems than humans [5]. Thus, social robots are helpful for JA interventions due to their social and physical characteristics [9].

Nevertheless, robot-mediated interventions are still an arguable topic due to divergent results [8]. In particular, some studies have shown opposite conclusions regarding the robot's ability to interact with children [10], [11]. These studies state that once children lose their focus on the robot or the therapy, it is hard to recover it merely using robot actions [10], [11]. Furthermore, studies use a wide range of social robots. Among these robots are the humanoid SARs CommU [12], Kaspar [13] and NAO [14]. The objective of those robotic implementations ranges from validating whether robots can help develop this ability [15] to creating a database, such as the orientation of the head, the gaze and the position of the body [14]. There are three guides for developing JA: the gaze, the verbal indication, and pointing at the target. However, they can be combined in multiple ways [16]. Additionally, some studies compare robotic and human intervention between typically developed children and children with ASD [17], [18]. Being a human-like robot is essential to representing social cues in real scenarios, such as facial distribution and pointing arms [12], [14].

In this context, this work studies the effects of combining therapist and robot participation during JA training in children with ASD. The robot's participation progressively increased throughout multiple sessions. The robot managed all the instructions in the last session with each child. This study also compares conventional JA interventions. The aim is to determine the best Therapist-Robot mediation relation in JA therapies for children with ASD. The study involved eight sessions in one month, attending two sessions per week with 16 children with ASD, using a system with RGB-depth cameras and facial recognition modules to estimate several JA indicators.

Fig. 1: Illustration of the ONO robot with active arms. (a) ONO's frontal view. (b) Mechanical design of the arm. (c) Inner view of the actual arm.

II. METHODOLOGY

An 8-session protocol was designed to assess robot-mediated JA interventions in individuals with ASD.

A. The ONO robot

This study uses the human-like ONO robot to mediate therapeutic interventions, developed by the OPSORO open-source project (Ghent University, Belgium) [19]. The actuation interface consists of 13 servomotors to move the mouth, eyes, and eyebrows. This robot can play sounds using an onboard speaker. A Raspberry Pi board controls the robot and provides a user-friendly web application. Ramirez *et al.* reported that active arms were necessary to provide target pointing and human characteristics, so the ONO's mechanical structure was modified [6]. Figure 1 illustrates the robot's appearance after the modifications for a more humanoid appearance and behavior.

B. Experimental Protocol

1) Participants Recruitment: This study was approved by the ethical committee of the Colombian School of Engineering Julio Garavito. The inclusion criteria were: (1) children who were attending therapies at the Howard Gardner Rehabilitation Clinic (Bogotá D.C., Colombia), (2) children diagnosed with ASD, (3) verbally-able children, and (4) children aged between 2 and 10 years old. The exclusion criteria were: (1) children with other cognitive disorders and (2) children with hearing or visual impairments.

An initial group of 16 children was selected. The children were formally recruited, and the caregivers provided their signed informed consent, allowing participation and the recording of the sessions. The group was divided into two groups by the health professionals. These groups were made, considering the age and functionality level of the children. The functionality level was assessed according to the Functional Independence Measure for Children (WeeFIM) [1]. This scale evaluates the functionality according to 18 items, scoring each one with an integer value between 1 and 7. These items are divided into three categories: (1) self-care, (2) mobility skills, and (3) cognition [1]. For this study, only cognitive skills were assessed for each child, where the maximum value was 35.

Both groups were randomly divided into the Control Group (CG) and the Robot-Assisted Group (RAG). Table I summarizes the division of both groups (CG and RAG), the cognitive score and the age of each participant.

TABLE I: Summary of demographic and cognitive information of the participants. Eight children formed each group. Cognitive Score from 0 to 35.

ID	Control Group (CG)		Robot-Assisted Group (RAG)	
	Cognitive Score	Age [y.o.]	Cognitive Score	Age [y.o.]
1	32	9	30	10
2	21	6	24	8
3	19	6	15	9
4	15	7	15	5
5	12	6	12	8
6	10	9	12	5
7	10	5	8	7
8	5	5	5	4
	15.50 ± 8.43	6.63 ± 1.60	15.13 ± 8.22	7.00 ± 2.14

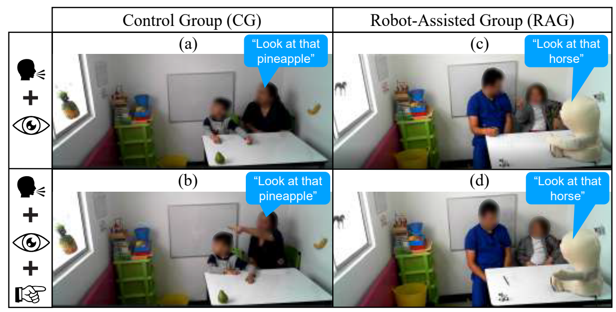

Fig. 2: Illustration of the two stimulus types and instructions during the Joint Attention (JA) activity. (a) Verbal and looking for CG. (b) Verbal, looking and pointing for CG. (c) Verbal and looking for RAG. (d) Verbal, looking, and pointing for RAG.

2) Session Environment and Measurement System: This study was conducted in a specially equipped consulting room at the Howard Gardner Rehabilitation Clinic. The facility for the tests consisted of two rooms: (1) A room for intervention with the children and (2) a room for remote control, monitoring of the robot, and measuring systems. To set eye-catching targets for the JA tasks, the session's room was equipped with three toy-like objects (Obj1, Obj2, and Obj3). Figure 2 illustrates examples of both groups' experimental setups.

To estimate JA indicators and assess performance, the Multimodal Environment for Robot-based intervention (MERI) developed by *Ramirez-Duque et al.* was used [6]. This system is equipped with a desktop CPU with an NVIDIA GeForce GTX 1080 graphic card and an i7-8086K processor. The CPU was configured to run a Linux operating system distribution, providing support for the Robotic Operating System (ROS) framework. The system integrates multiple RGB-Depth cameras to track users' faces, estimating the head pose, eye gaze, and users' field of view (FoV).

3) Session Procedure: Each child was asked to participate in a total of 8 sessions, attending two sessions per week. Each session consisted of several JA tasks, where the therapist or the robot instructed the children to look at specific objects in the room. To avoid target learning by the children, three

TABLE II: Description of the type of object and the number of instructions performed by the therapist and the robot at each session and group (Control Group (CG) and Robot-Assisted Group (RAG)).

Session	Type of Object	CG Therapist	RAG ONO	RAG Therapist
0	Vehicles	-	0	6
1	Animals	6	0	6
2	Fruits	6	1	5
3	Vehicles	6	2	4
4	Animals	6	3	3
5	Fruits	6	4	2
6	Vehicles	6	5	1
7	Animals	6	6	0

groups of objects were used: (1) vehicles, (2) animals, and (3) fruits.

Each session consisted of 6 instructions. The number of instructions performed by the therapist and the robot progressively changed over the sessions (See Table II). In the first sessions, the therapist requested most of the instructions, and in the last sessions, the ONO robot asked most of them. With the Robot-Assisted Group (RAG), an additional session was conducted at the beginning without the robot to determine the participants' initial performance.

The sessions were divided into three parts: (1) Introduction, (2) JA tasks, and (3) Feedback (See Figure 3). During the introduction, the therapist or the robot initiates the interaction with the children by talking to them and introducing the new objects in the consulting room. During the second and central part of the session, the JA instructions are given to the child following Table II distribution. Two behaviours were used for giving the instructions: (1) verbal communication and target looking, and (2) verbal communication, target looking, and target pointing. Regarding the target looking by the robot, it has no range of mobility in the neck. Thus, it only moved the eyes towards the object. After giving each instruction to the children, the therapist waited 3 seconds for the response. If the subject does not answer, the indication is repeated. This waiting time was aimed at analyzing the participant's behaviour.

Finally, the third part of the session consisted of giving neutral or positive feedback to the children according to their performance during the second phase of the JA tasks. All interactions with the participants were designed according to the Discrete Trial Teaching (DTT) procedure. The DTT is a widely used methodology in children with ASD for new behaviours and skills teaching [20]. This methodology is composed of three phases: (1) Antecedent Stimulus (AS), which consists of giving one or more cues to the child to generate a response; (2) Acceptable Response (AR), which evaluates if the child's response is the desired, and (3) the Reinforce Stimulus (RS), that depending on the response, good, neutral or no feedback is given to the child [20] (See Figure 3).

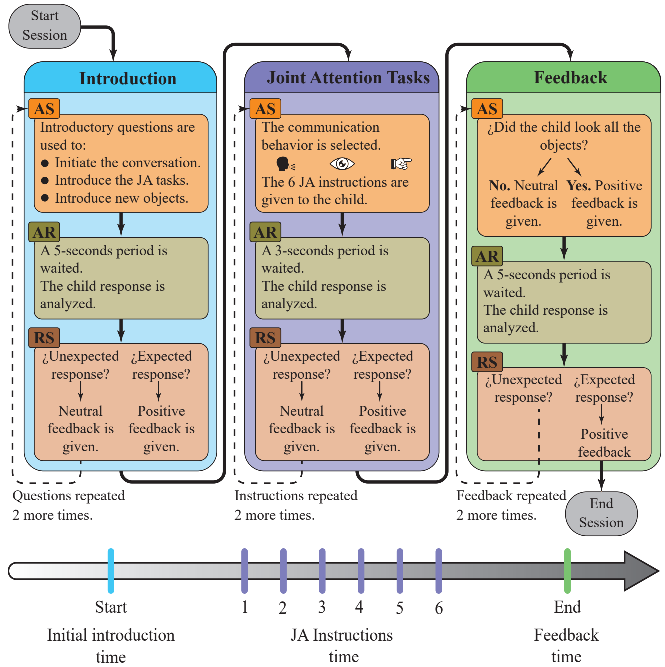

Fig. 3: Session's timeline and implementation of the (AS) Antecedent Stimulus, (AR) Acceptable Response, and (RS) Reinforcement of the Stimulus Discrete Trial Teaching methodology in the three main parts of the interventions.

C. Data Processing and Joint Attention Metrics

An illustration of the MERI outputs is shown in Figure 4. Additionally, each session time was obtained in seconds, as well as the II, JA task, and FF times to calculate the following metrics:

1) Active Metrics of JA:
   - Joint Attention Score (JAS): It evaluates the visual JA performance during the Joint Attention Activity (JAA). Two points were given each time the subject looked at the respective object after being asked the first time. If the child looked at the target on the second or third attempt, one point was given. Zero points were given if the child did not look at the object. A session's maximum Joint Attention Score (JAS) was 12 [9], [21].
   - II Visual Focus (VF_{II}): It represents the percentage of time that the child was looking at the therapist (VFT_{II}) and the robot (VFR_{II}), during the initial introduction (II) phase [9], [22].
   - JAA Visual Focus (VF_{JAA}): It represents the percentage of time that the child was looking at the therapist (VFT_{JAA}) and the robot (VFR_{JAA}), during the joint attention activity (JAA) [9], [22].
2) Passive Metrics of JA:
   - Verbal Engagement Score (VES): It measures the child's responses during the Initial Introduction (II) and the Final Feedback (FF) phases. Two points were given if the participant showed a desired answer on the first attempt. One point if additional attempts were required, and zero points if the child did not show the desired response. The maximum score for a session was 8.

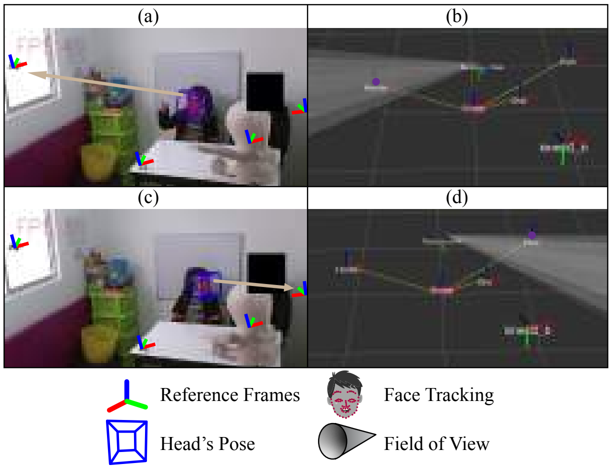

Fig. 4: Outputs of MERI. (a) Face tracking and eye gaze looking at an object on the window. (b) FoV measurement looking at the object on the window. (c) Face tracking and eye gaze looking at the object on the wall. (d) FoV measurement looking at the object on the wall.

- Feedback-seeking Score (FSS): It consists of giving 1 point, each time that the child turns back to see the robot (FSSR) or the therapist (FSST), after succeeding on a JA task. Thus, 6 was the maximum value for the FSSR and FSST [9].

Afterwards, the FoV cone was used to extract the visual measurements (FSS_T, FSS_R, VFR_R, VFR_T, VFT_{JAA}, and VFR_{JAA}) and the JAS. However, there were some cases when the MERI system could not estimate the FoV due to problems in face tracking (e.g., occlusions). Hence, these cases were analyzed by two researchers and two psychologists from the videos. The VES was obtained by analysing the videos when the therapist and ONO verbally interacted with the participant, according to the expected responses. The researchers carried out this process.

To get general results and determine how the variables evolved, data were clustered regarding the number of sessions according to the respective group. Finally, statistical tests were performed to determine significant differences between the interactions with the therapist and the robot in the CG and RAG interventions. The Kolmogorov-Smirnov test was applied to both groups' samples in each session (i.e., the data set was not normally distributed). Thus, all the implemented statistical tests were non-parametrical.

III. Results and Discussion

A. Active JA metrics

Table III contains the average time each session took in the interventions for both groups. It can be seen that the RAG took more time than the CG because careful coordination between the therapist and the robot operator was required. The total time was not representative to identify performance or human-robot interaction insights in the therapy. Hence, the VFT_{II} and VFT_{JAA} are presented as percentages of the time expended in the whole session to compare the time metrics between groups.

TABLE III: Average time of the session parts (Initial Introduction (II), Joint Attention (JA) Task, Final Feedback (FF)).

Group	II Time (s)	JA Task Time (s)	FF Time (s)
CG	28.81 ± 13.47	41.90 ± 19.58	12.12 ± 6.70
RAG	34.63 ± 17.24	58.09 ± 24.75	14.27 ± 5.85

TABLE IV: Results of the Wilcoxon Rank Sum test between the robot-assisted group (RAS) and the control group (CG) for all the metrics in session 0.

Metrics	RAG	CG	p-value
JAS (score 0-12)	6.62 ± 3.42	7.00 ± 3.29	0.66
$FSST$ (score 0-6)	0.25 ± 0.46	0.37 ± 0.51	0.99
VFT_{JAA} (%)	10.00 ± 5.85	8.44 ± 4.97	0.24
VFT_{II} (%)	13.12 ± 3.94	16.12 ± 4.85	0.16
VES (score 0-8)	3.87 ± 1.55	4.00 ± 1.41	0.96

To determine that both groups were homogeneous, the Wilcoxon Rank Sum (WRS) test was executed for all the metrics in the initial session (See Table IV). No p-value was lower than 0.05. Thus, it can be considered that both groups were homogeneous at the beginning of the study.

Figure 5 displays the JAS results. Given that different participants conform to the RAG and CG, a Mann-Whitney test was implemented to determine significant differences among the sessions. As shown in Figure 5, only the last two sessions showed substantial differences. Also, it can be seen how the median values of the JAS in the CG are, in general, increasing. In contrast, the JAS values decrease from session 4 in the RAG. Nevertheless, a continuous increment is presented from sessions 0 to 3, with the highest scores in session 3 for the RAG.

According to the results, the JAS score level matches the cognitive level of the participants. Also, some participants presented the worst JAS performance when ONO was leading the majority stimulus, while another part could still follow the instructions. These results are concordant with previous studies that have reported that JA interventions using only a robot as an intermediary do not improve the children's JA performance. This is because of the robot's limitations regarding recovering children's attention and the number of stimuli it can provide [10], [11].

Regarding the visual focus, the outcomes can be seen in Table V, showing the time percentage of the participant's visual focus on the therapist (RAG_VFT, for RAG and CG_VFT, for CG) and the ONO (RAG_VFR). The results are divided into the Joint Attention Activity and the Introduction. The Wilcoxon Signed-Rank (WSR) test was also implemented to compare the visual focus between the RAG and the CG. Significant differences were found for comparisons in all sessions ($p < 0.01$). In general, the percentage of time the participants were looking at the robot was always higher than that of the therapist. Considering the setup implemented for the RAG, having more time for visual contact with the robot could be because the robot was on the table in front of the participant. At the same time, the therapist was next to him. So, the children's field of

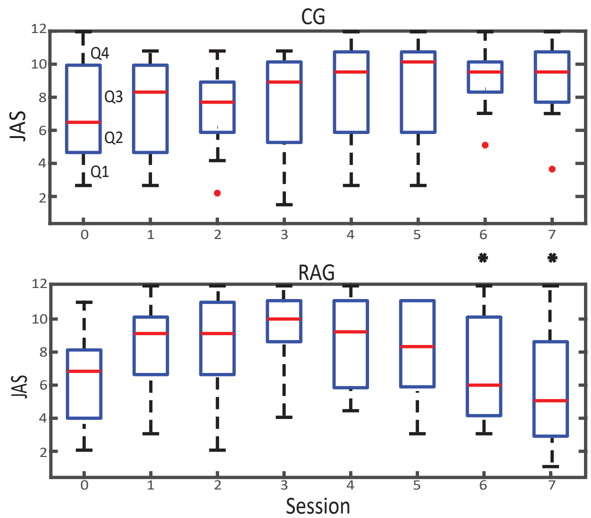

Fig. 5: Box-and-whisker diagrams for JAS. Control Group (CG), and Robot-Assisted Group (RAG) (* $p < 0.05$).

view includes the robot without exerting any effort, change of position, or movement. Moreover,

Furthermore, the average time percentages were higher in the Introduction section compared to the JAA section (See Table V). Such differences are related to the type of interaction in each phase. In the JAA, children had to move their visual focus to find and see the targets. Thus, they looked at the therapist or the robot, even if they did not want to. In the Introduction section, participants were encouraged to look at the interlocutor (the robot or the therapist). Considering that ONO always leads the introduction section in the RAG sessions and is crucial for JA interventions, these results reflect that the robot can keep the participants' visual focus. Besides, according to the literature, studies have shown that children with ASD are more attracted to interact with robots [5] and primarily to interact visually [23].

B. Passive JA Metrics

Table VI shows the feedback-seeking score results related to the therapist (RAG_FSST) and the robot (RAG_FSSR) in RAG, and the scores related to the therapist in CG (CG_FSST). It can be seen that RAG_FSSR means are always higher than the other groups. Significant differences were found for all sessions between RAG and CG for the feedback-seeking score ($p < 0.05$). Except for session 2, significant differences were found between RAG_FSST and RAG_FSSR. Additionally, Table VI also shows the results of the verbal engagement score for the questions asked by the therapist (CG_VES) and the robot (RAG_VES). No significant differences were found between both groups for all sessions.

In general terms, the therapist interacted verbally with the children during the Introduction section. Therefore, it was expected to have the highest VFT values during this part. However, although the VES results display that the children answered some questions effectively, they did not make eye contact during this interaction. On the other hand, the low

TABLE V: Results for visual focus time during the introduction and joint attention activity. Visual focus on the robot ($RAG_V FR$) and on the therapist $RAG_V FT$, $CG_V FT$). $+$ differences between RAG and CG. $*$ differences between $RAG_V FR$ and $RAG_V FT$.

Session	RAG_VFR	RAG_VFT	CG_VFT
JAA Visual Focus time [%]			
1^{+*}	33.12 ± 16.01	13.00 ± 10.84	5.00 ± 2.78
2^{+*}	30.00 ± 14.99	14.00 ± 8.02	4.88 ± 3.00
3^{+*}	35.25 ± 9.91	11.38 ± 7.56	4.25 ± 3.81
4^{+*}	36.88 ± 8.95	8.38 ± 6.52	4.38 ± 3.34
5^{+*}	29.13 ± 11.22	6.00 ± 4.04	2.00 ± 1.85
6^{+*}	30.75 ± 14.09	4.88 ± 3.80	2.13 ± 2.03
7^{+*}	32.13 ± 13.07	6.13 ± 5.08	2.13 ± 1.96
Introduction Visual Focus Time [%]			
Session	RAG_VFR	RAG_VFT	CG_VFT
1^{+*}	49.88 ± 23.84	10.13 ± 7.81	19.00 ± 5.92
2^{+*}	36.63 ± 20.65	12.13 ± 5.00	18.25 ± 5.92
3^{+*}	51.00 ± 26.78	11.38 ± 6.50	14.13 ± 8.08
4^{+*}	55.13 ± 26.23	8.13 ± 5.94	14.88 ± 8.84
5^{+*}	52.88 ± 20.97	10.00 ± 6.19	20.88 ± 8.87
6^{+*}	37.88 ± 21.68	11.00 ± 8.05	12.88 ± 4.55
7^{+*}	45.00 ± 22.00	11.13 ± 6.85	14.88 ± 5.03

values for the VFT_{JAA} are concordant with the low $FSST$ results (RAG: 0.250 and CG: 0.375) and the JAA results (RAG: 6.625 and CG: 7.000). These results show that the children looked at the corresponding target after being asked to. However, they seldom turned to look for visual feedback after completing the task. This could mean the child knows the answer and does not need the support and congrats from the mediator, or the child is not interested in the visual feedback. This is a social limitation related to ASD [9].

One limitation of this study is the lack of an external agent during the CG sessions to compensate for the influence of the robot's presence. In CG sessions, the therapist was the only agent for JA stimulus. Regarding the distribution of instructions given by the therapist and the robot, this study suggests that the best outcomes come from the sessions when the therapist and the robot have the same quantity of functions during the activity. This also confirms that a robot is a tool for the therapist and not a replacement agent which can assist JA therapies alone [10].

Finally, as mentioned, the best JAA scores were obtained in the sessions where the therapist and the robot participated equivalently, but the other JAA metrics were always higher for the robot, regardless of the session number. This work concludes that social robots that represent human characteristics are valuable for eliciting JA skills. However, the therapist's intervention is required to support the therapies and the robot instructions and keep the children focused.

IV. CONCLUSIONS & FUTURE WORK

The study conducted in this work determined the most effective distribution of the use of SAR in JA interventions, considering the limitations obtained in the application as the setup (the RAG had two mediators, while the CG had only one), the range of motion of the ONO robot (head

TABLE VI: Results for feedback-seeking to the robot (RAG_FSSR) and the therapist (RAG_FSST, CG_FSST), as well as results for verbal engagement in the robot-assisted group RAG_VES and the control group (CG_VES). + differences between RAG and CG. * differences between RAG_FSSR and RAG_FSST.

Feedback Seeking Score [%]			
Session	RAG_FSSR	RAG_FSST	CG_FSST
1^{+*}	2.75 ± 1.28	0.88 ± 0.64	0.75 ± 0.46
2^{+}	2.00 ± 1.20	0.88 ± 0.64	0.63 ± 0.52
3^{+*}	2.88 ± 1.17	0.88 ± 0.85	0.75 ± 0.46
4^{+*}	2.75 ± 1.58	0.63 ± 0.74	0.88 ± 0.35
5^{+*}	2.13 ± 0.99	0.63 ± 0.52	0.75 ± 0.46
6^{+*}	2.50 ± 1.51	0.38 ± 0.52	0.63 ± 0.52
7^{+*}	2.25 ± 1.58	0.50 ± 0.54	0.38 ± 0.52
Verbal Engagement Score [%]			
Session	RAG_VES	CG_VES	-
1	3.50 ± 2.62	4.13 ± 2.10	-
2	4.13 ± 1.96	4.00 ± 1.69	-
3	4.12 ± 2.17	3.62 ± 2.38	-
4	3.63 ± 1.76	4.25 ± 1.83	-
5	4.13 ± 2.30	4.88 ± 1.81	-
6	3.50 ± 1.93	4.63 ± 1.60	-
7	4.38 ± 2.07	4.25 ± 1.98	-

movement), and the difference between the two groups due to the comparison with traditional therapy and not with an equivalent one in terms of the number of mediators, the contribution was not directly compared concerning the control group. However, the significant differences between conventional therapy provided at the Howard Gardner Clinic and that applied with only one mediator were evaluated. The results showed that, although the CG presents an improvement in JA task performance progressively throughout the sessions, the robot in the RAG encourages participants to perform other passive and active JA actions, such as looking at and talking to the person they are verbally interacting with, seeking visual feedback after being instructed to look at an object, and paying more visual attention during JAAs.

However, the purpose of the study, which was mainly based on finding the correct and efficient combination between instructions given by the robot and instructions given by the therapist, the expected result was obtained, where the highest score was obtained in different metrics and between both groups when the robot and the therapist provide the same number of prompts in the session. Therefore, this work suggests that the therapist can use social robots to train JA tasks and obtain more advanced JA behaviours with this equal distribution.

Finally, in future work, it can be said that a low-cost robot that can perform the necessary head movements for this activity should be chosen. Although it was not a limitation in the results, it should be selected to evaluate the difference that this feature can bring to the study. In addition, assess the difference with an equivalent control group and not with traditional therapy to compare both groups and obtain conclusive results regarding the impact obtained with the robotic implementation in JA with the most efficient therapist-robot distribution found in this study.

REFERENCES

[1] J. Baio *et al.*, "Prevalence of autism spectrum disorder among children aged 8 Years - Autism and developmental disabilities monitoring network, 11 Sites, United States, 2014," *MMWR Surveillance Summaries*, vol. 67, no. 6, 2018, ISSN: 15458636.

[2] M. Carpenter *et al.*, "Interrelations among Social-Cognitive Skills in Young Children with Autism," *Journal of Autism and Developmental Disorders*, vol. 32, no. 2, pp. 91–106, 2002, ISSN: 01623257.

[3] R. Jordan, *Autistic Spectrum Disorders*. David Fulton Publishers, 2013, ISBN: 9780203827352.

[4] C. Kanai *et al.*, "Social skills in autism spectrum disorders," in *Handbook of social behavior and skills in children*, Springer, 2017, pp. 217–248.

[5] B. Scassellati, "How social robots will help us to diagnose, treat, and understand autism," in *Robotics research*, Springer, 2007.

[6] A. A. Ramírez-Duque *et al.*, "Robot-Assisted Autism Spectrum Disorder Diagnostic Based on Artificial Reasoning," *Journal of Intelligent and Robotic Systems: Theory and Applications*, 2019, ISSN: 15730409.

[7] E. Kim *et al.*, "Bridging the Research Gap: Making HRI Useful to Individuals with Autism," *Journal of Human-Robot Interaction*, pp. 26–54, 2012.

[8] M. Begum *et al.*, "Are Robots Ready to Deliver Autism Interventions? A Comprehensive Review," *International Journal of Social Robotics*, vol. 8, no. 2, pp. 157–181, 2016, ISSN: 18754805.

[9] A. A. Ramırez-Duque *et al.*, "Robot-assisted intervention for children with special needs: A comparative assessment for autism screening," *Robotics and Autonomous Systems*, p. 103 484, 2020.

[10] B. Scassellati *et al.*, "Improving social skills in children with asd using a long-term, in-home social robot," *Science Robotics*, vol. 3, no. 21, eaat7544, 2018.

[11] Z. Zheng *et al.*, "A randomized controlled trial of an intelligent robotic response to joint attention intervention system," *Journal of Autism and Developmental Disorders*, pp. 1–13, 2020.

[12] H. Kumazaki *et al.*, "The impact of robotic intervention on joint attention in children with autism spectrum disorders," *Molecular autism*, vol. 9, no. 1, pp. 1–10, 2018.

[13] S. E. Mengoni *et al.*, "Feasibility study of a randomised controlled trial to investigate the effectiveness of using a humanoid robot to improve the social skills of children with autism spectrum disorder (kaspar rct): A study protocol," *BMJ open*, 2017.

[14] E. Billing *et al.*, "The dream dataset: Supporting a data-driven study of autism spectrum disorder and robot enhanced therapy," *PloS one*, vol. 15, no. 8, e0236939, 2020.

[15] A. Di Nuovo *et al.*, "Deep learning systems for estimating visual attention in robot-assisted therapy of children with autism and intellectual disability," *Robotics*, vol. 7, no. 2, p. 25, 2018.

[16] D. O. David *et al.*, "Developing joint attention for children with autism in robot-enhanced therapy," *International Journal of Social Robotics*, vol. 10, no. 5, pp. 595–605, 2018.

[17] R. Simut *et al.*, "Can you cure me? children with autism spectrum disorders playing a doctor game with a social robot," *International Journal on School Health*, vol. 3, no. 3, 2016.

[18] Z. E. Warren *et al.*, "Can robotic interaction improve joint attention skills?" *Journal of autism and developmental disorders*, vol. 45, no. 11, pp. 3726–3734, 2015.

[19] C. Vandevelde *et al.*, "The Use of Social Robot Ono in Robot Assisted Therapy," Tech. Rep.

[20] S. Eikeseth *et al.*, "Discrete trial teaching and discrimination training," in *Handbook of early intervention for autism spectrum disorders*, Springer, 2014, pp. 229–253.

[21] P. G. Esteban *et al.*, "How to build a supervised autonomous system for robot-enhanced therapy for children with autism spectrum disorder," *Paladyn*, vol. 8, no. 1, pp. 18–38, 2017, ISSN: 20814836.

[22] S. Lemaignan *et al.*, "Artificial cognition for social human–robot interaction: An implementation," *Artificial Intelligence*, vol. 247, pp. 45–69, 2017.

[23] S. M. Anzalone *et al.*, "Quantifying patterns of joint attention during human-robot interactions: An application for autism spectrum disorder assessment," *Pattern Recognition Letters*, 2019.